

OFFLINE · rendering-to-localization map reconstruction

1 Gaussian scaffold



1.20M frozen RGB splats

2 Feature tracks



real mapping observations

3 Triangulated anchors



independent PnP geometry

4 Candidate universe



48K localization evidence

OFFLINE · topology and descriptor reconstruction

5 Compact topology



7,145 single-descriptor anchors

6 Self-localization



keep · swap · miss · attractor

current map

matching outcomes update
descriptor reconstruction

7 Sparse deployment



SuperPoint → top-1 → one PnP/RANSAC

ONLINE · one-shot sparse deployment